# Kexuan Zhang

PhD Researcher · Graphics and Systems Programmer
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## Education and Research

**University of Leeds**, UK 2025 – Present

*PhD in Perceptual Graphics*
Research: virtual reality, foveated rendering and high-performance graphics.
Focus: stereo reprojection and image inpainting.

**University of Leeds**, UK 2023 – 2024

*MSc in High-Performance Graphics and Games Engineering, Distinction*
Average: 74.7. Dissertation: real-time global illumination with Light Propagation Volumes.

**Jianghan University**, Wuhan, China 2017 – 2021

*BEng in Computer Science and Technology (Software Engineering)*
Average: 84.3/100; rank: 18/212.

## Software Engineering Experience

**Shanghai Zihao Information Technology Co., Ltd** Jul. 2022 – May 2024

*Game Server Development Engineer*

- Developed high-performance .NET game server systems and investigated performance bottlenecks and MongoDB integration.
- Built Unity skill-pipeline tools, Excel configuration tooling and Jenkins CI/CD pipelines.
- Developed Roslyn-based static analysis tools to enforce code consistency.

**Shanghai Zihao Information Technology Co., Ltd** Dec. 2021 – Jun. 2022

*Intern Programmer*

- Built experience with C#, Win32, multithreading, networking and Unix/Linux systems.

## Selected Projects

**VRI – Cross-platform Render Hardware Interface** Active development

*C ABI, C++23, Vulkan, D3D12, Metal, WebGPU, OpenGL* Personal project

- Develop an explicit rendering API with command buffers, descriptor sets and synchronization, paired with a header-only C++23 RAII wrapper.
- Implement backend capability queries, validation and portable Slang shaders. Browser examples and a Vulkan path tracer demonstrate the API; feature coverage is documented per backend.

**Light Propagation Volumes** 2024

*C++, OpenGL, Real-time Global Illumination* MSc dissertation

- Implemented an LPV renderer on VGFW to explore real-time indirect lighting.
- Integrated cascaded shadow maps, HBAO, screen-space reflections, FXAA, ACES tone mapping and bloom into the rendering pipeline.

**libvultra – Graphics Research Framework** 2024 – Present

*C++23, Vulkan, Slang, Render Graphs* Personal project

- Develop the current VRI-based framework for small rendering demos and research experiments on Windows and Linux.
- Build reusable rendering, shader and asset components; the early-stage development branch includes a glTF viewer, render-graph construction and shader compilation/reload.

**Cells of Division – High Path Games** 2024 – Present

*Co-founder and Programming Lead; Godot* Independent game

- Design the program architecture and core gameplay subsystems for a bullet hell roguelike; lead the programming team and coordinate development.
- Built framework systems including ECS, configuration, localisation and buffs. A free demo is available on Steam.

**OmniLyrics – Cross-platform Lyrics Tool** Active development

*C#, .NET 10, Avalonia, CLI/TUI* Personal project

- Built a shared .NET core for desktop, terminal and status-bar interfaces, with bilingual karaoke lyrics and platform playback integrations.
- Publish Windows, macOS and Linux releases, including signed and notarized macOS GUI applications.

## Technical Skills

- Languages: C/C++, C#, Python, Java and Lua. Currently studying CUDA independently through CIS5650.
- Graphics and engines: Vulkan, OpenGL, Slang, render graphs, global illumination, Unity and Godot.
- Systems and tooling: multithreading, networking, .NET performance, Git, XMake/CMake, Jenkins, Roslyn and MongoDB.
- Languages spoken: Mandarin (native); English (IELTS 6.5).

## Publication

Li, S. and **Zhang, K.** (2021). Mobile remote video surveillance system. *Computer Systems and Applications*, 30(06), 82–87. doi:10.15888/j.cnki.csa.007939. Second author.

## Honors and Awards

National First Prize, Bochuang Cup AI Competition 2019, 2020
Provincial Second Prize, Blue Bridge Cup Programming Competition 2018, 2019
Provincial Third Prize, Programming Ladder Competition 2018, 2019